Institutional breadth as a cross-sectional signal: research memo

2026-11-08

This memo summarises the full technical report (`report/report.html`), which documents the data pipeline, the software, the design and every result in detail.

1. Question and signal

Does the change in the number of institutions holding a stock, read from Form 13F filings only after they are public, predict the cross-section of returns of liquid US stocks, net of costs, and does it add to momentum? The signal F1 is the change in breadth of ownership of Chen, Hong and Stein (2002, p. 180): among filers present in both quarters, holders at q minus holders at q-1, divided by the number of filers. It was chosen because it is defined in a peer-reviewed paper, needs only holdings counts (robust to value errors in 13F data), and connects to three mechanisms: gradual accumulation by institutions as information diffuses (an underreaction and momentum channel), informed institutional trading (Sias, Starks and Titman 2006), and short-sale constraints (Chen, Hong and Stein). The threat is that the signal was public for up to 45 days before formation, and that Nagel (2005) found the breadth effect reversed after 1998, before our sample starts.

2. Data and design

SEC Form 13F data sets (2013Q2 to 2026Q2) give holdings; Tiingo gives total-return daily prices for live and dead tickers; OpenFIGI maps CUSIPs to permanent identifiers. A `targets` pipeline moves data through bronze (as delivered), silver (point-in-time tables) and gold (signals and panels), with a data-quality suite before each pin.

- **Point in time.** A filing is usable the trading day after its filing date; amended reports are versioned so a reader at date t sees only the versions current at t, and a late report enters at the first formation after it is public. A random sample of 300 filings checked against EDGAR's acceptance times confirms that every one was public before the close of its usable date, when the portfolio trades.
- **Universe.** At each formation (end of February, May, August, November), the 1,000 most liquid US common stocks with a price above USD 5, a year of history, trading on 90% of recent days and a confirmed identity.
- **Portfolios.** F1 quintiles, rebalanced quarterly and drifting in between, under three weightings: cs (signal times liquidity, pre-registered), equal weight and risk parity. The spread is the top minus the bottom quintile.
- **Evaluation.** FF5 plus momentum (FF6) alpha with Newey-West standard errors, rank IC per formation, a double sort within momentum terciles, a monotonicity test, terminal-loss stress, and costs of 10 to 50 bps per dollar traded plus 100 bps borrow. A credible signal must pass under all three weightings.

3. Results

Spread	Mean, % a year	FF6 alpha	t	95% CI
F1, cs (pre-registered)	4.48	2.25	1.06	[−1.93, 6.42]
F1, equal weight	3.10	1.61	0.94	[−1.74, 4.95]
F1, risk parity	2.46	0.61	0.41	[−2.28, 3.50]
F1 within momentum terciles	3.39	2.38	1.44	[−0.86, 5.63]
F1, terminal stress	4.88	2.67	1.26	[−1.48, 6.83]
F1 net, 25 bps + 100 bps borrow	0.71	−1.81	−0.84	[−6.02, 2.40]
Momentum (reference)	5.22	1.19	0.39	[−4.72, 7.10]

1. **Predictive content: no evidence.** The cs spread earns 4.48% a year gross, with an FF6 alpha of 2.25% and an interval of [−1.93, 6.42]. The mean rank IC is 0.011 (t = 0.74). Returns are flat from the first to the fourth quintile and rise only at the fifth; the monotonicity test does not reject a flat relation (p = 0.53).

2. **Economic viability: not viable.** Net of 25 bps and 100 bps borrow, the cs spread earns 0.71% a year and the other weightings earn less. F1 changes almost completely each quarter (rank autocorrelation 0.11), so each leg trades its full value at every rebalance; the break-even fee is 40 bps.
3. **Incremental to momentum: not shown.** F1 earns a positive spread in every momentum tercile, averaging 3.39% a year, but the interval includes zero, and the F1 and momentum spreads have a correlation of 0.86.

The weighting hurdle is not met: the spread is positive under all three schemes but no alpha is significant.

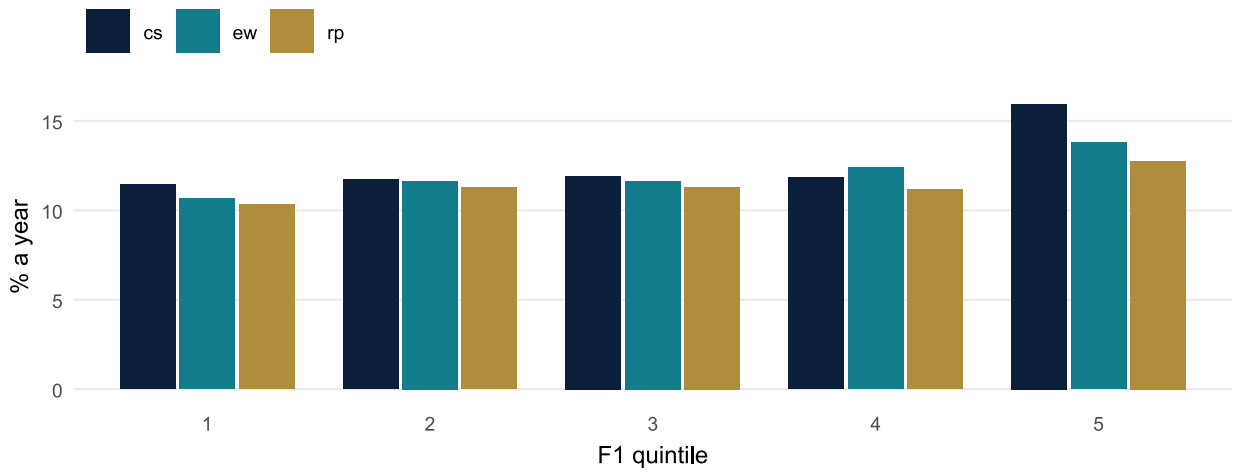


Figure 1: Annual return of each F1 quintile (1 = lowest breadth change).

The extreme quintiles hold more liquid stocks than the middle ones even with equal weights (liquidity percentile 0.55 and 0.64 against about 0.44): the extremes of F1 are widely held stocks, which can gain or lose hundreds of holders in a quarter. At the last formation the three largest F1 values were INTC, MU, AMD, adding 708 to 863 holders each.

3.1. Costs

Weights	Fee, bps	Gross, % a year	Trading cost	Net
cs	0	4.48	0.00	4.48
cs	10	4.48	1.11	3.37
cs	25	4.48	2.77	1.71
cs	50	4.48	5.55	−1.06
ew	0	3.10	0.00	3.10
ew	10	3.10	1.19	1.91
ew	25	3.10	2.98	0.13
ew	50	3.10	5.95	−2.85
rp	0	2.46	0.00	2.46
rp	10	2.46	1.22	1.23
rp	25	2.46	3.05	−0.60
rp	50	2.46	6.11	−3.65

Costs absorb most of the gross spread because F1 has almost no persistence: the rank correlation of F1 between consecutive formations averages 0.11, so most of each quintile is replaced every quarter. Turnover is a property of the signal, not of the rebalancing rule. A borrow fee of 100 bps a year on the short leg costs a further 1 percentage point. Averaging F1 over two quarters should keep more of each quintile and cut the cost in proportion.

Much of the F1 return is a momentum return: the two gross spreads have a correlation of 0.86, which is why the FF6 alpha, which controls for momentum, is about half the raw spread. A 50/50 combination has a Sharpe ratio of 0.25, between momentum alone (0.21) and F1 alone (0.29): F1 is a less volatile way to hold a similar bet rather than a separate source of return. This comparison is descriptive and gross of costs.

4. How to read the result

The test has little power. With 152 months and a spread this volatile, an alpha would have to be about 10.8% a year to be detected four times out of five; the pre-registered threshold of economic interest was 2%. Momentum, the closest known anomaly, was itself insignificant in the window: UMD averaged 2.66% a year (t = 0.76) against 7.25% since 1963. The evidence is therefore inconclusive, not a rejection, and it is consistent with Nagel's finding that the breadth effect weakened after 1998.

UMD	Mean, % a year	t	CAPM alpha	t
study window	2.66	0.76	6.00	2.05
since 1963	7.25	3.98	8.36	4.96

5. Design choices and deviations

The hypotheses, definitions and decision rules were fixed before any return was computed; every later change is dated in the memo's deviation log and the report. The main ones:

- **Identity at t.** A CUSIP is accepted only if most holders report a value per share close to Tiingo's close at quarter end, recomputed at each formation.
- **Suspected CUSIP changes.** A stock where more than half of at least 20 holders are new is excluded for that formation: corporate actions that move holders to a new CUSIP otherwise look like mass initiations.
- **Denominator.** F1 divides by the matched cohort rather than all funds; ranks within a formation are identical.
- **Traded-day presence.** The universe requires trading on 90% of the previous 63 days, so that a stock whose vendor price stays frozen after a takeover, delisting or halt cannot enter it.
- **Short-leg weighting.** Under cs the bottom quintile is weighted by conviction in the short direction (negated F1 times liquidity).

6. Reproducibility

The project is an R package with a `targets` pipeline: `renv::restore()` then `targets::tar_make()` rebuilds every pin, the report and this memo. Each layer pins only after its data-quality suite passes; 300 unit tests run on every change and in continuous integration on a clean machine. The backtest engine, factoRverse, ships in the repository.

7. Limitations

Short sample and low power; a structural size tilt in F1; identity from today's OpenFIGI and Tiingo mappings (0.45% of 13F value cannot be priced, mostly dead firms); formation up to two weeks after most filings are public; stocks that stop trading after formation earn zero until the next formation. The size tilt matters most: F1 partly ranks stocks by how widely they are held.

8. Next steps

F1b (ownership change by value) and F1c (new holders relative to the stock's own holders) measure the same idea without the size tilt, each as a separate trial with false discovery rate control and a deflated Sharpe ratio. A costs study would test smoothing F1 over two quarters, which should raise persistence and cut turnover. Leave-one-year-out and sector checks would test whether the top-quintile premium is concentrated in a few episodes.